

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2022

Course Code: PHT100

Course Name: ENGINEERING PHYSICS A

PART A

	<i>Answer all questions. Each question carries 3 marks</i>
Q1	Comment the statement "smaller the damping, larger will be the relaxation time and greater the quality factor".
ans	$Q = \frac{\omega'}{2\gamma}$ From this equation it is clear that as the damping decreases Quality factor increases. Quality factor is inversely proportional to damping factor. hence smaller the damping, larger will be the relaxation time and greater the quality factor
Q2	State the laws of vibrations of a stretched string.
ans	$v = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$ Frequency of transverse vibrations of stretched string is directly proportional to square root of Tension and inversely proportional to square root of linear mass density. frequency is inversely proportional to length of the string.
Q3	Explain the formation of different colours in thin films with example.
ans	A thin transparent film observed in white light appears coloured Condition for darkness is $2\mu t \cos r = n\lambda$ Depends on Wavelength, Angle of refraction, Thickness At any region of film, if μ, t and $\cos r$ satisfies this condition for particular wavelength λ, that colour will be absent in that reflected light. Hence that part of the film will appear to have the colours of remaining light. It depends on angle of refraction r hence on angle of incidence i. So when we observe the same film from different positions the colour of the film will be varying. If the thickness is different, colour of the film will be varying. When t is very small, path difference between reflected rays $\lambda/2$. ie, darkness condition. Hence the film appears as dark.
Q4	What is diffraction? Compare diffraction pattern with interference pattern.
ans	Diffraction of light is the phenomenon of bending of light round the edges of an obstacle OR Encroachment of light into the geometrical shadow of the obstacle.

	Interference Diffraction Interference bands are formed by superposition of waves from two coherent sources Diffraction bands are formed by superposition of waves from different parts of the same wavefront Bands are of equal width Bands are of unequal width Bands of minimum intensity are almost dark Bands of minimum intensity are not dark Intensity of bright bands is same Intensity of bright bands is not the same.
Q5	Write the expression for energy of a particle trapped in one dimensional box and find the energy of first excited state.
ans	$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$ $E_n = n^2 h^2 / 8mL^2$ The different values of energy for n are called Eigen values. Since n is restricted, the particle cannot have all the values of energy. The quantity n is called quantum number.
Q6	What is meant by quantum dot and quantum wire?
ans	Quantum wires A nano wire or quantum wire is a wire of dimensions of the order of nanometers. Quantum wire is a one dimensional structure. Quantum wires are formed when 2 dimensions of the system are confined ie by making a thin wire of semiconductor. Two direction have quantum confinement. It is only free to move along one dimension that is along the wire. A quantum wire is an electrically conducting wire in which quantum effects influence the transport properties. Such effects appear in the dimension of Nano meters. So they are called Nano wires. Electrical conductance of wire is found to be quantized in multiples of $2e^2/h$. quantum wires can be used as an electron wave guide. It is used to make high speed lasers. Quantum dots Quantum dots are zero dimensional structure in which the electron is confined in all three dimensions. Their energy states are quantized in all three directions. consider a tiny dot of semi conductor material then we have a 3D quantum confinement of charge carriers. That is there are no freely moving electrons or holes in any direction .This is called “quantum dot”. Although this quantum dot consists of thousands of atoms it behaves like a single atom with discrete energy levels rather than many atoms. When the size of the crystal become so small that it approaches the size of excitonbohr radius, then the electron energy levels will change from continuous to discrete values. That is their is a finite separation between energy levels. The energy gap increases with the decreases in the diameter
Q7	State Ampere’s circuital theorem. Give its mathematical statement.

ans	AMPERE'S CIRCUITAL LAW The line integral of the magnetic flux density (B) for a closed path is equal to μ times the net current enclosed by the path $\oint_C \underline{B} \cdot d\underline{l} = \mu_0 I_{encl}$
Q8	Define Divergence of a vector function. Give its physical significance.
ans	The divergence is an operator, which takes in the vector-valued function defining this vector field, and outputs a scalar-valued function measuring the change in density of the fluid at each point. divergence is a vector operator that operates on a vector field producing a scalar field giving the quantity of the vector field's source at each point. More technically, the divergence represents the volume density of the outward flux of a vector field from an infinitesimal volume around a given point. As an example, consider air as it is heated or cooled. The velocity of the air at each point defines a vector field. While air is heated in a region, it expands in all directions, and thus the velocity field points outward from that region. The divergence of the velocity field in that region would thus have a positive value. While the air is cooled and thus contracting, the divergence of the velocity has a negative value.
Q9	Write a note on high temperature superconductors with two examples.
ans	High Temperature Superconductors are ceramic superconductors with a high transition temperature greater than 40 K. In low temperature superconductors it is very difficult and expensive to maintain low temperature for a very long period. It is very difficult to maintain the liquid helium temperature (4.2K). nowadays it is possible to replace the expensive liquid He with cheaper and more efficient coolant liquid nitrogen. The temperature T_c can be raised using microwaves or laser radiations. In ceramic superconductors, the cell contains 1 atom of rare earth metal, 2 Barium atoms, 3 copper atoms and 7 Oxygen atoms. Since no. of atoms of metal are in the ratio 1:2:3, such ceramic superconductors are called 1-2-3 superconductors and it is working at T_c of 90K. Copper oxide superconductors belong to another group of HTSC working at T_c of 120K
Q10	Explain the working of a solar cell.
ans	A solar cell is an LED operating in the reverse method. Also called photovoltaic cell. Solar cell produces electrical energy from light energy. Sunlight is absorbed at the p-n interface. It excites electrons from the valence band of p-type to the conduction band of n-type. This leaves a hole in the valence band. Sunlight produces electron-hole pairs. The electric field at the junction pulls the electrons towards the n-region and holes towards the p-region. As a result a current flows from p-terminal to the n-terminal through an external circuit.

Application

It is applied in automobile shock absorbers door closer mechanism and recoil mechanism in gun. If the door closes without performing any oscillation, it is an example of critically damped system.

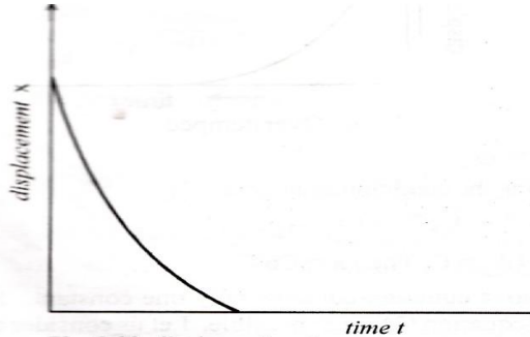


Fig. Critically damped

Case3: $\gamma < \omega$ The damping is low

Here $\sqrt{\gamma^2 - \omega^2}$ is imaginary

$$\text{Let } \sqrt{\gamma^2 - \omega^2} = i\omega' = i\sqrt{\omega^2 - \gamma^2}$$

$$\text{Where } \omega' = \sqrt{\omega^2 - \gamma^2} \text{ -----(8)}$$

From eq (3),

$$X = A_1 e^{(-\gamma + i\omega')t} + A_2 e^{(-\gamma - i\omega')t} \text{ -----(9)}$$

$$= e^{-\gamma t} (A_1 e^{i\omega' t} + A_2 e^{-i\omega' t})$$

$$= e^{-\gamma t} (A_1 (\cos \omega' t + i \sin \omega' t) + A_2 (\cos \omega' t -$$

$$i \sin \omega' t)$$

$$X = e^{-\gamma t} ((A_1 + A_2) \cos \omega' t + i(A_1 - A_2) \sin \omega' t)$$

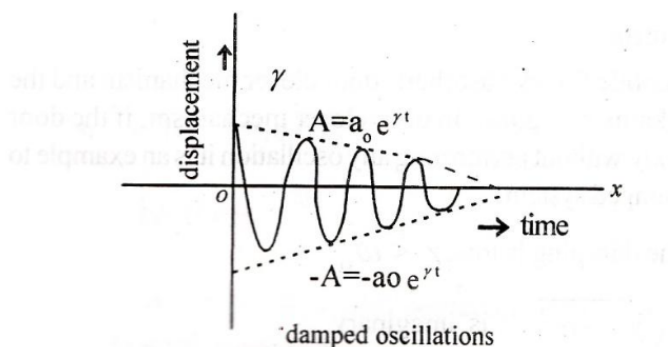
$$\text{Put } (A_1 + A_2) = x_0 \sin \varphi$$

$$\text{And } i(A_1 - A_2) = x_0 \cos \varphi$$

$$X = x_0 e^{-\gamma t} \sin(\omega' t + \varphi) \text{ ----- (10)}$$

Equation (10) shows that the motion is oscillatory. The amplitude $x_0 e^{-\gamma t}$ is not a constant but decreases with time. The motion is not periodic.

The oscillations are as shown in the figure.

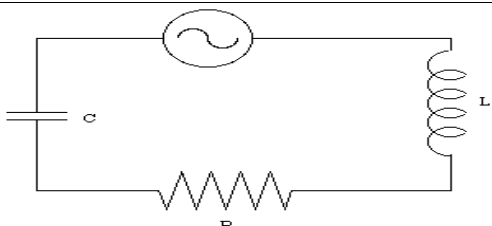
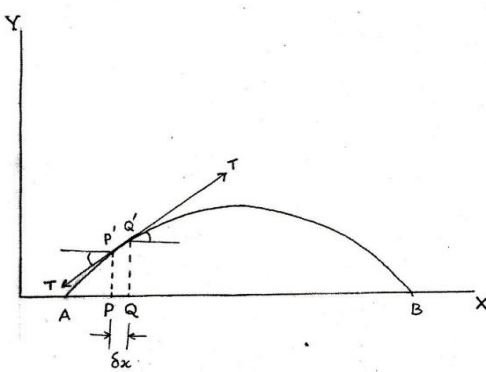


b Show that LCR circuit acts as an electrical analogue of mechanical oscillator.

ans **A pure LC circuit is an electrical analogue of simple pendulum. In the case of simple pendulum, the energy alternates between potential and kinetic energy.**
In the case of LC circuit when the electrical oscillation takes place, the energy is alternately shared in the capacitor as electric field and in the inductor as magnetic field.

The frequency oscillation in LC circuit,
$$n = \frac{1}{2\pi\sqrt{LC}}$$

Forced oscillation in a series LCR circuit

	Applying Kirchoff's law to the circuit $V_L + iR + V_C = V_0 \sin \omega t$ $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = V_0 \sin \omega t$ ie $\frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = V_0 \sin \omega t$  This is the differential equation in the case of forced oscillations in LCR circuit
Q12 a	Considering the transverse vibrations of a stretched string derive the one dimensional wave equation and show that velocity of the wave $v = \sqrt{T/\mu}$.
ans	Consider an infinitely long thin and uniform string stretched between two points by a constant tension T along the X-axis. Let the string be slightly displaced aside along the Y-axis and released. Transverse vibrations are set up in the string. Consider a small element PQ of the string of length δx. The magnitude of tension in the string will be same everywhere since the string is perfectly flexible. The tension T acts tangentially at every point on the string. P'Q' is the displaced position of the string.  The tension at P' and Q' acts along the tangent making angles θ_1 and θ_2 with the horizontal. Resolving the tension along x-axis and y-axis, Then net force on PQ acting X and Y direction are $F_x = T \cos \theta_2 - T \cos \theta_1$ $F_y = T \sin \theta_2 - T \sin \theta_1$ For small oscillations θ_1 and θ_2 are very small. $\cos \theta_1 = 1 \quad ; \quad \cos \theta_2 = 1$ Also $\sin \theta_1 = \tan \theta_1 \quad ; \quad \sin \theta_2 = \tan \theta_2 \quad (\tan \theta_1 = \sin \theta_1 / \cos \theta_1 = \sin \theta_1)$ Then $F_x = 0$ $F_y = T \tan \theta_2 - T \tan \theta_1$ So the net force acting on the element δx in the displaced position is only along Y-axis. $F_y = T(\tan \theta_2 - \tan \theta_1)$ If μ is the mass per unit length of the string (linear density), mass of the element δx, $m = \mu \delta x$ $\text{Acceleration} = \frac{\partial^2 y}{\partial t^2}$ According to the Newton's second law of motion $F = ma$ $F = \mu \delta x \frac{\partial^2 y}{\partial t^2}$

$$T(\tan \theta_2 - \tan \theta_1) = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$\text{But } \tan \theta_1 = \left(\frac{\partial y}{\partial x} \right)_x ; \quad \tan \theta_2 = \left(\frac{\partial y}{\partial x} \right)_{x+\delta x}$$

$$\text{or } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\tan \theta_2 - \tan \theta_1}{\delta x} \right]$$

$$\text{ie } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right]$$

when $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right] = \frac{\partial^2 y}{\partial x^2} \quad \text{--- (5)}$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} \quad \text{--- (6)}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}$$

This is the differential equation of a vibrating string.

Comparing this equation with the standard wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

$$\frac{1}{v^2} = \frac{\mu}{T}$$

$$v^2 = \frac{T}{\mu} \quad v = \sqrt{\frac{T}{\mu}}$$

This is the velocity of transverse wave on a stretched string.

$$v = v \lambda v = \frac{v}{\lambda} v = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$$

Frequency of the standing wave with n antinodes is given by

$$v = \frac{n}{2l} \sqrt{\frac{T}{\mu}}$$

Law of transverse vibrations of a stretched string

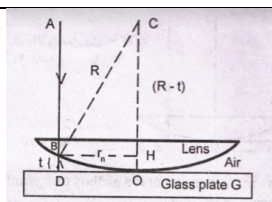
Frequency of transverse vibrations of stretched string

$$v = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

		- Fundamental frequency is directly proportional to the square root of Tension - Fundamental frequency is inversely proportional to the length of vibrating string - Fundamental frequency is inversely proportional to the square root of linear density
b		The displacement of a wave is represented by $y = 0.25 \times 10^{-3} \sin(0.025x - 500t)$, where x is measured in metres and t in secs. Deduce amplitude, time period, velocity and wavelength?
ans		$y = 0.25 \times 10^{-3} \sin(0.025x - 500t)$ $y = A \sin(kx - \omega t)$ $A = 0.25 \times 10^{-3}$ $k = 0.025$ $\frac{2\pi}{\lambda} = 0.025$ $\lambda = \frac{2\pi}{0.025} = 251.3 \text{ m}$ $\omega = 500$ $2\pi f = 500$ $f = \frac{500}{2\pi} = 79.57 \text{ Hz}$ $T = \frac{1}{f} = 0.01255$ $v = f\lambda$ $= 19995.9 \text{ m/s}$

Module 2

Q13 a	Explain the construction and band formation in Newton's ring apparatus. Derive an equation to find the wavelength of incident monochromatic light.
ans	The arrangement consists of a plano convex lens of large radius of curvature placed on an optically plane glass plate. A thin film of air of varying thickness is formed between the lens and the glass plate. The thickness of thin film is zero at the point of contact and gradually increases towards the edge of the lens. A beam of monochromatic light is incident normally on the lens. A part of the light is reflected from the top surface of the thin film another part of light is reflected from the top surface of the glass plate. These two reflected beams interfere destructively or constructively and produce dark or bright band. If a point appears dark, all the points along a circle through this point are dark since the thickness of air film is same along a circle. So we get a dark ring. If the point appears bright, we get a bright ring. Thus alternate dark and bright rings of increasing radii are observed. As the radii increase, the rings become thinner and closer. These are called newton's rings.



At the point of contact between the lens and the glass plate, the thickness of air film $t=0$. Hence the path difference $2t=0$. But at the time of reflection, the reflected wave from the glass plate undergoes a phase change of π (path difference of $2t$). So the waves at the centre are interfering destructively and hence a dark spot is obtained at the centre of Newton's ring.

Let C be the centre of curvature of the curved surface of the lens and R be its radius of curvature.

Let r_n be the radius of dark ring, $BH=r_n$

The condition for the formation of the n th dark ring at D is,

$$2t \cos r = n\lambda \quad \dots\dots(1) \text{ where } n = 0, 1, 2, 3, \dots$$

Since the light is incident normally, $r = 0$ or $\cos r = 1$.

Also $\lambda = \lambda$ for air.

$$\text{i.e., } 2t = n\lambda \quad \dots\dots(2)$$

From the right angled triangle CBH,

$$CB^2 = CH^2 + BH^2$$

$$\text{i.e., } R^2 = (R-t)^2 + r_n^2$$

$$= R^2 + t^2 - 2Rt + r_n^2$$

Since t is very small t^2 is neglected.

$$R^2 = R^2 - 2Rt + r_n^2$$

$$r_n^2 = 2Rt \quad \dots\dots(3)$$

$$2t = n\lambda \text{ from (2)}$$

$$\lambda r_n^2 = Rn\lambda \quad \dots\dots(4)$$

$$\text{Or } r_n = \dots\dots(5)$$

WAVELENGTH OF LIGHT

The radius of n th dark ring is

$$r_n =$$

$$r_n^2 = Rn\lambda$$

If D_n is the diameter of the n th dark ring, then

$$D_n^2 = 4Rn\lambda \quad \dots\dots(9)$$

If D_{n+k} is the diameter of the $(n+k)$ th dark ring, then

$$D_{n+k}^2 = 4R(n+k)\lambda \quad \dots\dots(10)$$

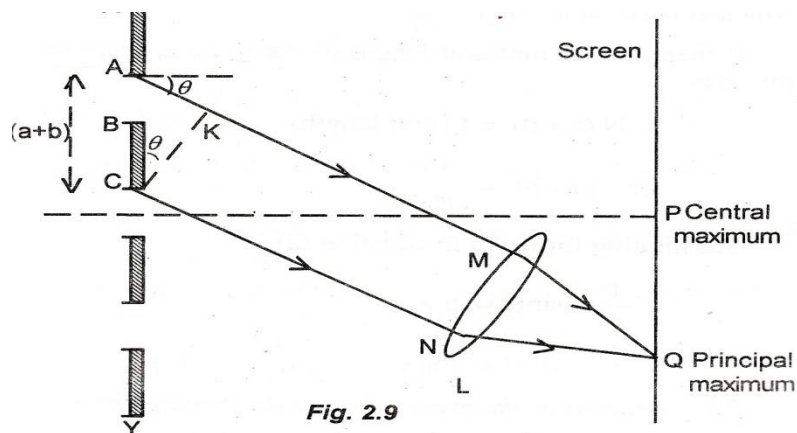
$$\text{Then } D_{n+k}^2 - D_n^2 = 4kR\lambda$$

b

What is an antireflection coating? Find the minimum thickness of antireflection coating (magnesium fluoride of refractive index 1.42) to be deposited on a glass plate of refractive index 1.5, for light of wave length $5800\text{\AA}$.

ans	$f = \frac{\lambda}{4\mu_g} = \frac{5800 \times 10^{-10}}{4 \times 1.42} = 1.02112 \times 10^{-7} \text{ m}$ $= \underline{\underline{0.102 \mu\text{m}}}$
Q14 a	Explain resolving power of a grating? Explain the formation of spectra by a plane transmission grating by deriving grating equation.
ans	PLANE TRANSMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point . The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light. Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces . Usually a grating has 5000 to 12000 lines per cm. Original grating is highly expensive and it is difficult to construct. Hence their photographic replicas are used. A thin layer of collodion solution is spread uniformly on the original grating and then it is allowed to harden. Such a collodion film with the impression of the ruling is kept in between two optically plane glass plates. This is the transmission grating that we use There are two important conditions for a good quality grating. 1. The number of lines per cm must be very large. 2. The spacing between the lines must be equal. Consider a plane transmission grating placed perpendicular to the plane of paper. AB represents a slit and BC represents a line. Let 'a' be the width of each slit and 'b' be the width of each line. The distance (a+b) is called grating element or grating constant. A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit. Each point of the wave front sends out secondary waves in all direction. The straight and parallel waves from each point can be focussed on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum. The position of central maximum is same for all the wavelength. The central maxima will have the same colour as the incident light.

Consider two waves diffracted from two points A and C of slit. They travel along AM and CN. Draw CK perpendicular to AM. Then the path difference between the two waves is AK.



From triangle ACK

$$\sin \theta = AK/AC$$

$$AK = AC \sin \theta$$

$$AK = (a+b) \sin \theta$$

$$\text{If } (a+b) \sin \theta = n\lambda \dots\dots(1)$$

where $n=0,1,2,3,\dots$ two waves interfere constructively. This is called principal maximum.

For different values of n , there are different values of θ .

If $n=1$, it is the first order principal maximum, if $n=2$, it is the second order principal maximum and so on. Thus on either side of central maximum, a number of principal maxima are obtained.

If there are N lines/unit length of the grating, there are N slit also.

$$N(a+b) = 1 \text{ (unit length)}$$

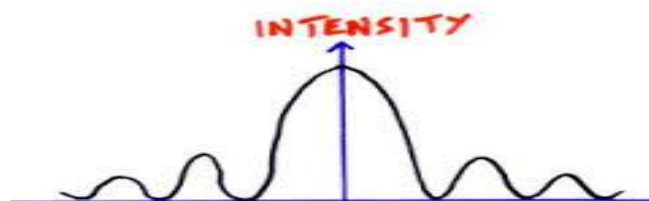
$$(a+b) = 1/N$$

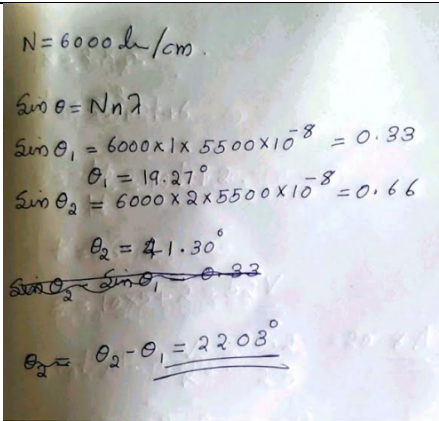
Substitute in equation (1)

$$1/N \sin \theta = n\lambda$$

$$\sin \theta = Nn\lambda$$

This is known as grating law or grating equation.



		For a grating θ is different for different colours (λ), for each value of n . If white light is used, it gets split up into different colours.	
	b	Consider a plane transmission grating having 6000 lines/cm. Find the angular separation between first and second order maxima when a light of wavelength $5500\text{\AA}$ is incident normally on it.	
	ans		
Module 3			ans
	Q15 a	Derive time dependent Schrodinger equation for a particle moving in a potential $V(x)$. Hence derive time independent Schrodinger equation.	
	ans	SCHRODINGERS WAVE EQUATION FOR A FREE PARTICLE AND TIME DEPENDENT EQUATION A particle in motion is associated with a wavefunction that contains the information about the motion. A plane progressive wave that propagates along x-direction is given by $\Psi = Ae^{i(kx - \omega t)} \quad \text{.....(1)}$ Where k is the wave vector. $k = 2\pi/\lambda$ ω is the angular frequency Einstein's formula for photon energy is $E = h\nu = \hbar\omega$ De-Broglie's expression for matter wavelength is $\lambda = \frac{h}{p}$ $p = \frac{h}{\lambda} = \frac{h}{\frac{h}{mv}} = mv$	

$$\frac{h}{2\pi} = \hbar k$$

Using the above expression, equation (1) becomes

$$\Psi = A \exp(i(\hbar kx - \hbar \omega t))$$

$$\Psi = A \exp(i(px - Et))$$

On partial differentiation of ψ with respect to x , twice, we get

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi \dots \dots \dots (2)$$

Differentiating w.r.t time,

$$\frac{\partial \psi}{\partial t} = -iE \frac{\psi}{\hbar} \dots \dots \dots (3)$$

These are equivalent to

$$\frac{p^2 \psi}{2} = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} \dots \dots \dots (4)$$

$$= -i\hbar \frac{\partial}{\partial x} (-i\hbar \frac{\partial}{\partial x}) \psi$$

$$\text{From 3 and 4, } p = -i\hbar \frac{\partial}{\partial x}$$

This is called momentum operator

	$E\psi = i\hbar \frac{\partial \psi}{\partial t} \quad \text{.....(5)}$ $E = i\hbar \frac{\partial}{\partial t} \quad \text{.....(5a)}$ This is energy operator For a free particle total energy is given by $E = \frac{p^2}{2m}$, since $V=0$ Ie, $E\psi = \frac{p^2}{2m} \psi$(6) Using equations (4) and(5), Equation 6 becomes $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = i\hbar \frac{\partial \psi}{\partial t} \quad \text{.....(7)}$ This is Schrodinger's equation for a
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According to Schrodinger, de-Broglie's wavelength holds good for any particle moving in any field of force with potential energy V.

Then total energy = Kinetic energy + Potential energy

$$E = \frac{1}{2}mv^2 + V$$

$$= \frac{p^2}{2m} + V$$

$$\text{ie, } p^2 = 2m(E - V)$$

$$\text{Or } p = [2m(E - V)]^{1/2}$$

The wave equation in Cartesian co-ordinate system can be written as

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{u^2} \frac{\partial^2 \psi}{\partial t^2} \dots\dots\dots(9)$$

$$\text{Where } \psi = \psi_0 e^{-i\omega t} \dots\dots\dots^{*(10)}$$

u is the velocity of motion of wave and $\psi(x, y, z, t)$ represents the amplitude with the particle.

From (10), we get

$$\frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi$$

So Equation (9) becomes

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = -\frac{\omega^2}{u^2} \psi \dots\dots\dots(11)$$

$$\frac{\omega^2}{u^2} = \frac{(2\pi\nu)^2}{(\nu\lambda)^2} = \frac{4\pi^2}{\lambda^2}$$

So we have

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = -\frac{4\pi^2}{\lambda^2} \psi \quad \text{or}$$

This is a general equation which is independent of time.

Now introduce the concept of de-Broglie wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E - V)}}$$

$$\lambda^2 = \frac{h^2}{2m(E - V)}$$

Substituting the wavelength in equation (12)

$$\nabla^2 \psi + \frac{4\pi^2}{h^2} [2m(E - V)] \psi = 0$$

$$\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

This represents Schrödinger's time-independent wave equation.

	b	State and explain Heisenberg's uncertainty principle. Explain natural line broadening mechanism on the basis of this principle.
	ans	According to this principle it is impossible to have an accurate measurement of the position and momentum of particles simultaneously. or The Uncertainty Principle states that the simultaneous determination of the exact position and momentum of a moving particle is impossible The product of uncertainty (error) in the measurement of position of the particle (Δx) and the uncertainty in the momentum (ΔP_x) of the particle is of the order of Planks constant. $\Delta x \cdot \Delta P_x = \hbar$ ($\hbar = h/2\pi$) 3. Natural line broadening The average time period that takes between the excitation of an atom and the time it radiates is 10^{-8}. Thus the photon energy is uncertain by the amount $\Delta E = \hbar / \Delta t$ $= 1.055 \times 10^{-26}$ J It means that the excited energy levels have a finite energy spread. Thus the energy levels of the atom must have a finite width or in other words the spectral lines can never be sharp but have some broadening in frequency, called natural line broadening.
	Q16 a	What are nanomaterials ? Discuss the mechanical optical and electrical properties of nano materials.
	ans	Electrical properties Ionization potential at nano sizes are higher than that at bulk solids. The change in electrical properties is not the same in all nano materials. In nano ceramics the electrical conductivity increases with the decrease in particle size. But in metals it decreases. In the case of bulk metals the energy level of electron is

		continuous where as in nano sized metals the energy levels are discrete. Optical properties Depending on the particle size, same substance shows different colours. Gold spheres of 100 nm size appears orange while that of 50 nm size appears green The particles can be made to emit or absorb specific wavelength of light by varryingit's size Mechanical properties Nano crystalline nickel is as strong as hardened steel. Because of nano size many mechanical properties such as hardness, elastic modulus, fracture toughness, scratch resistance and fatigue strength are modified 30-50 % lower elastic moduli than conventional 2-7 times higher hardness Super plastic behaviour in brittle ceramics
	b	Give any six applications of nanomaterials
	ans	<u>Nanomedicine</u> ● Cell imaging ● Treating cancer ● For diagnostic purposes ● The drug consumption and side effects can be reduced significantly by targeted medicine. ● This is done depositing the medicine in the affected region only and in the correct doze ● NT helps to reproduce or repair damaged tissue. In Energy Applications in the field of energy are 1. By reducing energy consumption by designing LEDs or Quantum Caged Atoms(QCAs) etc. 2. By increasing energy production efficiency by designing specific catalysts to improve the combustion efficiency in IC engines 3. By using more environmentally friendly energy systems such as fuel cells powered by hydrogen chemistry and Environment Chemical catalysts from nanoparticles are highly useful Filtration : Nanofiltration helps 1. Waste water management Information and communication Memory storage devices such as Nano-RAM is developed using nanotubes

		Food Textiles Clothes made of nanofibers are stain-repellent and wrinkle-free These textiles can be washed less frequently
Module 4		
	Q17 a	Explain the terms (i)Magnetization (ii)Magnetic permeability (iii)Relative permeability and (iv) Susceptibility. Derive the relation between magnetic susceptibility and relative permeability.
	ans	INTENSITY OF MAGNETISATION (M) It is a measure of the magnetisation of a magnetised specimen. It is defined as the magnetic moment per unit volume of the material. M= But Moment =m*2l MAGNETIC SUSCEPTIBILITY The magnetic susceptibility is a measure of how much a material become magnetised in an applied magnetic field. It is defined as the ratio of magnetization M to the applied magnetizing field intensity H. $\chi = \frac{M}{H}$ MAGNETIC PERMEABILITY The magnetic permeability is defined as the property of the material to allow the magnetic lines of force to pass through it. The permeability of the material is equal to the ratio of the flux density B in the medium to the field intensity H. $\mu = \frac{B}{H}$ RELATIVE PERMEABILITY The relative permeability of the material is the comparison of the permeability with the permeability of air or vacuum Relative permeability $\mu_r = \frac{\mu}{\mu_0}$

		The flux density B in the magnet is $B = \mu_0(H + M)$ $B = \mu_0 H + \mu_0 M$ $\frac{B}{H} = \frac{\mu_0 H}{H} + \frac{\mu_0 M}{H}$ $\mu = \mu_0 + \mu_0 \chi$ $\mu = \mu_0(1 + \chi)$ $\frac{\mu}{\mu_0} = (1 + \chi)$ $\mu_r = 1 + \chi$
	b	The magnetic flux through a closed circuit with resistance 2.5Ω varies with time obeying the equation $\phi = 5t^2 + 2t + 6$. What will be the induced emf and current in it at time = 5 seconds.
	ans	$R = 2.5\Omega$ $\phi = 5t^2 + 2t + 6$ $\mathcal{E} = \frac{d\phi}{dt} = 10t + 2$ $\text{at } t = 5s$ $\mathcal{E} = 10 \times 5 + 2 = \underline{52V}$ $\text{Current } I = \frac{V}{R} = \frac{52}{2.5} = \underline{20.8A}$
	Q18 a	Starting from the field equations in free space, derive the velocity of electromagnetic waves in free space. How did Maxwell show that light is electromagnetic in nature?

7. Show that EM waves travels in freespace with the velocity of light

Maxwell's equations are

$$\nabla \cdot D = \rho$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

In vacuum,

$$J = 0, \rho = 0$$

$$\nabla \cdot D = 0$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial D}{\partial t}$$

$$D = \epsilon_0 E, B = \mu_0 H$$

$$\begin{array}{l|l|l|l} \nabla \cdot D = 0 & \nabla \cdot B = 0 & \nabla \times E = -\frac{\partial B}{\partial t} & \nabla \times H = \frac{\partial D}{\partial t} \\ \nabla \cdot \epsilon_0 E = 0 & \nabla \cdot \mu_0 H = 0 & \nabla \times E = -\frac{\partial}{\partial t} \mu_0 H & \nabla \times H = \frac{\partial}{\partial t} \epsilon_0 E \\ \nabla \cdot E = \frac{\sigma}{\epsilon_0} = 0 & \nabla \cdot H = \frac{0}{\mu_0} = 0 & \nabla \times E = -\mu_0 \frac{\partial H}{\partial t} \text{---(3)} & \nabla \times H = \epsilon_0 \frac{\partial E}{\partial t} \text{---(4)} \\ \nabla \cdot E = 0 \text{---(1)} & \nabla \cdot H = 0 \text{---(2)} & & \end{array}$$

Take curl on both sides of the equation 3

$$\nabla \times \nabla \times E = -\mu_0 \nabla \times \frac{\partial H}{\partial t}$$

$$\boxed{\nabla \times \nabla \times A = \nabla(\nabla \cdot A) - \nabla^2 A}$$

$$\nabla(\nabla \cdot E) - \nabla^2 E = -\mu_0 \frac{\partial}{\partial t} \nabla \times H$$

using eqn 1 and eqn 4

$$-\nabla^2 E = -\mu_0 \frac{\partial}{\partial t} \epsilon_0 \frac{\partial E}{\partial t}$$

$$\nabla^2 E = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} \text{---(5)}$$

Take curl on both sides of the eqn 4

$$\nabla \times \nabla \times H = \epsilon_0 \nabla \times \frac{\partial E}{\partial t}$$

$$\nabla(\nabla \cdot H) - \nabla^2 H = \epsilon_0 \frac{\partial}{\partial t} \nabla \times E$$

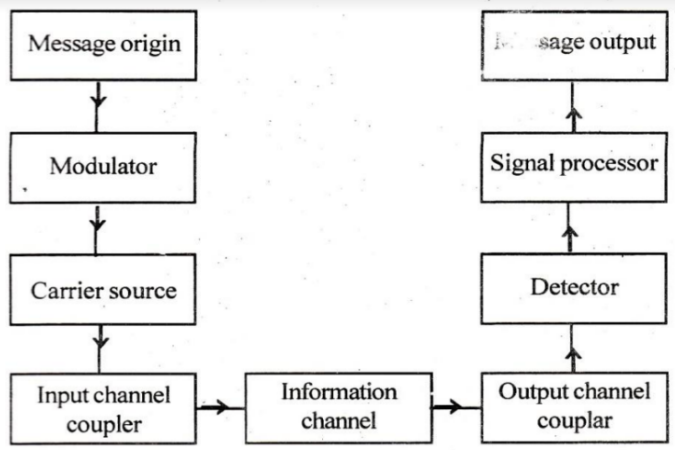
using eqn 2 and 3

$$-\nabla^2 H = \epsilon_0 \frac{\partial}{\partial t} \mu_0 \frac{\partial H}{\partial t}$$

$$\nabla^2 H = \epsilon_0 \mu_0 \frac{\partial^2 H}{\partial t^2} \text{---(6)}$$

		Comparing with the standard wave equation, $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ hence we can write $\frac{1}{v^2} = \mu_0 \epsilon_0$ $v^2 = \frac{1}{\mu_0 \epsilon_0}$ $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ Substituting the values of μ_0 $v = 3 \times 10^8 \text{ m/s.}$
	b	State and explain Poynting's theorem.
	ans	Poynting vector The magnitude and direction of flow of energy per unit area per unit time in an Electromagnetic wave travelling in freespace is represented by a vector known as Poynting vector. $S = E \times H$ unit is Watt per square meter. Poynting vector represents the instantaneous power density associated with the EM field at a given point.
Module 5		
	Q19 a	What is critical temperature and critical field? Write the relation between these two. Explain the conductivity of superconductors using BCS theory.
	ans	Transition temperature (T_c) is that temperature at which the material changes from its normal conducting state to superconducting state. It is found that superconducting transition is reversible. Above the transition temperature, the material is in the normal state and below the transition temperature, it is in the superconducting state. Effect of magnetic field - Critical field (H_c) Superconductivity vanishes if a sufficiently strong magnetic field is applied.

	The minimum strength of a magnetic field required to destroy the superconducting nature of a metal at a particular temperature is called critical field H_c. When the applied field is below H_c material will be in superconducting state and if it exceeds H_c, it gets converted into its normal state. The variation of critical field H_c with temperature is represented by a parabolic curve as in fig. H_c is 0 at T_c and increases when the temperature decreases BCS THEORY Bardeen, Cooper and Schrieffer developed a new theory to explain superconductivity of materials and this is known as BCS theory. According to this theory, superconductivity is mainly due to the interaction of phonons and electrons. When an electron is advancing through a path in between 2 sets of positive ions in the lattice, electron attracts nearby positive ions and lattice gets distorted. This distortion produces a region of increased positive charge density and is called phonon. If a second electron approaches this region, this electron also attracted by the positive region. Thus these 2 electrons are held together by force of attraction by phonon. Thus 2 electrons are coupled together at low temperature and this pair is called cooper pair. Cooper pair behaves like single particle. Cooper pair has zero spin and zero linear momentum. At absolute zero almost all electrons are coupled together as cooper pairs Superconducting energy gap The net energy of electrons in a cooper pair is less than the total energy of these individual electrons. There is an energy gap in the energy spectrum of electrons. All the Cooper pairs are occupying lower energy states and energy gap prevents them from unpairing. The minimum energy required to break up cooper pairs is called superconducting energy gap. When temperature increases, energy gap decreases. At T_c energy gap becomes zero and cooper pairs are broken up and thus electrons are separated.
b	State and explain Meissner effect. Show that superconductors are perfect diamagnets.
ans	The property of expulsion of magnetic flux density from the interior of a superconducting material during the transition from normal state to superconducting state is called Meissner effect. When a specimen is placed in a magnetic field H, magnetic lines are passing through it. () $B = \mu_0 H$ Where,

		B = flux density inside the specimen μ_0 = Permeability M = magnetization When the specimen is cooled below the transition temperature, all magnetic lines are cancelled from the specimen. Since $B=0$, $() \ 0 \ 0 \ H \ M$ $H = -M$ Magnetic susceptibility $\chi = M/H = -1$ Hence the superconductor becomes a perfect diamagnet at the transition temperature.
Q20	a	Describe fibre optic communication system with a block diagram. List four advantages of fibre optic communication.
ans		BLOCK DIAGRAM AND FUNCTION OF EACH COMPONENT The message origin, modulator, carrier source, input channel coupler together constitute transmitter. Optic fibres are used as the information channel. Output channel coupler, detector, signal processor and message output constitute receiver.  <pre> graph TD MO[Message origin] --> M[Modulator] M --> CS[Carrier source] CS --> ICC[Input channel coupler] ICC --> IC[Information channel] IC --> OCC[Output channel coupler] OCC --> D[Detector] D --> SP[Signal processor] SP --> MO2[Message output] </pre> 1. MESSAGE ORIGIN Converts all non electrical messages to electrical signals using transducer. Eg; microphone converts sound energy to electrical energy 2. MODULATOR Imposing a message on a carrier wave for propagations is called modulation I. At first it converts electrical message into proper format. II. Secondly it imposes this format on a carrier wave for propagation.

There are 2 types of modulation. Analog and digital modulation. In analogue modulation message is transmitted in a continuous manner and in digital modulation message is transmitted in discrete manner with the help of binary digits. Digital modulation is preferred for long distance communication.

3. CARRIER SOURCE

Carrier source produces carrier waves on which the messages are transmitted. In fiber communication system light waves are the carrier waves. LED or Laser diodes are used to generate stable and monochromatic waves . The information is imposed on light waves.

4. INPUT CHANNEL COUPLER

This directs modulated light waves into the information channels.

In the case of radio or television broadcasting systems antenna delivers radio frequency waves into atmosphere for propagation. Here antenna behaves as input channel coupler

5. INFORMATION CHANNEL

This is a path to transmit the information from transmitter to receiver .Here very fine and long optic fibres are used as information channel. Modulated light signals are transmitted through optic fibre by principle of total internal reflection.

6. OUTPUT CHANNEL COUPLER

	This directs the modulated light signals from the information channel to the detector .Eg. Antenna in radio and television broadcasting systems. 7. DETECTOR This detects and separates the messages from the modulated signals. ie, demodulation takes place. Here light signals are converted into electric current using photo detector. 8. SIGNAL PROCESSOR This filters and selects the required frequency from waves. The selected frequency is amplified. The unwanted frequency is filtered out. 9. MESSAGE OUTPUT Here the original message is reproduced from the signals. The electrical pulses are converted into sound waves in the case of audio systems. ADVANTAGES OF COMMUNICATION WITH OPTIC FIBRES Extremely wide Bandwidth (Very high information carrying capacity) Smaller diameter, lighter weight cable Lack of cross talk between the parallel fibers. Immunity to electromagnetic interference Electric isolation Low cost Much safer than copper cables
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		Low transmission loss
b		The numerical aperture of an optic fibre is 0.38. If the difference in the refractive indices of the material of its core and the cladding is 0.05, calculate the refractive index of material of the core.
ans		$N.A = 0.38$ $n_2 - n_1 = 0.05 \quad \text{--- ①}$ $N.A = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 + n_2)(n_1 - n_2)}$ $0.38 = \sqrt{(n_1 + n_2) \times 0.05}$ $0.1444 = (n_1 + n_2) 0.05$ $2.888 = n_1 + n_2 \quad \text{--- ②}$ $\begin{array}{r} + \\ 0.05 \end{array} = \begin{array}{r} + \\ n_1 - n_2 \end{array} \quad \text{--- ③}$ $2.938 = 2n_1$ $n_1 = \frac{2.938}{2} = 1.469$